Dr. Marin Moran

Research Scientist in AI, Physics, and Rendering
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SUMMARY

I take methods from the physics of structured and disordered systems into machine learning and rendering, and I build instruments to measure what learned models actually know. That thread runs from symmetry-aware networks for crystal structures in my PhD, through a NeurIPS study finding no evidence that LLM confidence carries model-specific self-knowledge, to Stealthy Blue Noise in ACM Transactions on Graphics. At Electronic Arts SEED, I develop real-time generative AI for AAA games; I also design evaluation frameworks for scientific reasoning in frontier models.

EXPERIENCE

Electronic Arts (SEED), *AI, Physics, and Rendering Research Scientist*
Remote, Global Oct 2024 – Present
Real-time generative AI and LLM research for AAA games

- Designed neural architectures for real-time generative systems under production frame-time constraints, reducing memory use, register pressure, and ALU cost relative to deployed baselines without loss of output quality
- Technical lead for LLM integration in real-time interactive systems, spanning architecture, fine-tuning, evaluation, and production integration
- Led cross-functional research on LLM-driven interactive experiences from concept through working prototype
- Conceived and led Stealthy Blue Noise, a spectral construction for sparse rendering adapted from the physics of disordered systems, accepted to SIGGRAPH for publication in ACM Transactions on Graphics

Solomonoff Consultancy, *Founder & Principal Consultant*
Remote Feb 2024 – Present
Evaluation and verification for scientific reasoning in frontier AI systems

- Designed scientific-reasoning benchmarks for domains where ground truth is costly and binary correctness is insufficient
- Led the delivery of PhD-level evaluation datasets across physics, chemistry, and drug discovery, coordinating more than 100 domain experts
- Advised on alignment and deployment questions at the boundary of technical evaluation and responsible use

University of Liverpool, Materials Innovation Factory, *Post-Graduate Machine Learning Researcher*
Liverpool, UK Oct 2019 – Oct 2024
Symmetry-aware geometric deep learning for physical systems

- Designed Site-Net (Digital Discovery, 2023), a transformer that learns invariant representations of crystal structures, infinite periodic graphs with no canonical ordering, using global self-attention over real-space supercells to capture long-range interactions
- Established Deep InfoMax, which maximizes mutual information between local and global representations, as an effective self-supervised methodology for domains where reconstruction-based approaches are intractable (Digital Discovery, 2025)
- Scaled training across SLURM clusters using DeepSpeed, custom CUDA kernels, and symmetry-aware optimizations via PyTorch Geometric, and built bespoke data architectures for high-throughput multi-GPU pipelines
- Co-built a locally hosted cloud service for deploying supervised materials-informatics models, used by the Materials Innovation Factory research group

Electronic Arts, *Physics Software Engineer (Internship)*
Guildford, UK Sep 2023 – Mar 2024
Memory optimization and legacy code modernization for physics systems

- Redesigned the core data architecture of a major internal Frostbite development tool, halving its memory footprint in typical workloads (~5 GB saved)
- Built an in-engine documentation interface that surfaced the physics system's internals to non-specialist developers across the studio
- Designed and shipped a production agentic system on GPT-3 for legacy codebase modernization, combining LLM-driven refactoring with symbolic verification to keep changes safe under deployment

EDUCATION

University of Liverpool, *PhD, Deep Learning for Materials Science*
Liverpool, UK Oct 2019 – Sep 2024

- Doctoral thesis, An information oriented approach to materials informatics
- Conceived and led LLMs Show No Signs of Individuated Metacognition (NeurIPS), with Mark E. Whiting

University of Liverpool, *Integrated Masters (First Class Honours), Physics*
Liverpool, UK Sep 2015 – May 2019

- First Class Honours
- Focus on computational physics and machine learning applications

PUBLICATIONS

- **Stealthy Blue Noise**. **M. Moran**, W. Donnelly, A. Wolfe. *ACM Transactions on Graphics (SIGGRAPH)*.
- **Establishing Deep InfoMax as an effective self-supervised learning methodology in materials informatics**. **M. Moran**, M. W. Gaultois, V. V. Gusev, D. Antypov, M. J. Rosseinsky. *Digital Discovery*.
- **LLMs Show No Signs of Individuated Metacognition**. **M. Moran**, M. E. Whiting. *NeurIPS*.
- **Site-Net: using global self-attention and real-space supercells to capture long-range interactions in crystal structures**. **M. Moran**, M. W. Gaultois, V. V. Gusev, D. Antypov, M. J. Rosseinsky. *Digital Discovery*.
- **A database of experimentally measured lithium solid electrolyte conductivities evaluated with machine learning**. C. J. Hargreaves, M. W. Gaultois, L. M. Daniels, E. J. Watts, V. A. Kurlin, **M. Moran**, et al. *npj Computational Materials*.
- **The Liverpool materials discovery server: a suite of computational tools for the collaborative discovery of materials**. S. Durdy, C. J. Hargreaves, M. Dennison, B. Wagg, **M. Moran**, J. A. Newnham, M. W. Gaultois, M. J. Rosseinsky, M. S. Dyer. *Digital Discovery*.
- **Discovery of a Low Thermal Conductivity Oxide Guided by Probe Structure Prediction and Machine Learning**. C. M. Collins, L. M. Daniels, Q. Gibson, M. W. Gaultois, **M. Moran**, et al. *Angewandte Chemie International Edition*.

SKILLS

Languages & Frameworks. Python, C++, MATLAB, C#, Java, CUDA, JavaScript, PyTorch (Lightning), TensorFlow/Keras, JAX, Scikit-learn, PyTorch Geometric, Hugging Face, DeepSpeed, NumPy, Pandas, SciPy, Ray, XGBoost, ONNX

AI/ML Specializations. Large Language Models, Transformer Architectures, Deep Learning, Representation Learning, Generative AI (Diffusion Models, VAEs, GANs), Graph Neural Networks, Geometric Deep Learning, Contrastive Learning, Neural NLP, Computer Vision

Systems & Infrastructure. HPC (SLURM), Distributed Training, GPU Compute (CUDA Kernels), OpenMP, Graphics Programming (HLSL, GLSL), Real-time Physics Simulation, Game Engines (Unreal, Frostbite, Unity), Docker, Singularity, MLOps, Azure, Full Stack Development, Agentic Systems, MongoDB, SQLite, HDF5

Domain Expertise. Materials Science, DFT (Density Functional Theory), Molecular Dynamics, Crystallography, VASP, ASE (Atomic Simulation Environment), Frontier AI Safety, Scientific AI Verification, Real-time Game AI, Physics Simulation, Computational Chemistry, Low-level Systems (6502 Assembly, C++/CLI)

Game Development. Unreal, Unity, Frostbite, Perforce

INTERESTS

Creative Pursuits. Tabletop RPG design and community coordination, Vocal performance (classical and extended techniques), Retro game development (6502 assembly for NES)

Philosophy & Ethics. Philosophy of technology, Applied ethics and moral philosophy, History of ideas

Quantitative Hobbies. Market microstructure, Network dynamics, Complex systems modeling